Finlay Donald Webb

Contact Information

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Education

The University of Edinburgh, Edinburgh, UK, September 2023 - June 2027
B.Sc Mathematics (Hons)
Grade: Year 1, Year 2, Year 3 Sem 1 - First Class Honours

Research Experience

Extreme Properties of Randomly Oriented Networks
Supervisor: Jonathan Noel - University of Victoria,
June - August 2026 (Upcoming)

- **Overview**: Specific topic TBD

Robustness of Clique-Factors in Regular Graphs Above the Hajnal-Szemerédi Threshold
Supervisor: Shoham Letzter - University College London,
May 2025 - Present

- **Overview**: Investigated the robustness of K_r-factors in regular graphs, focusing primarily on the triangle case ($r = 3$) to develop tractable approaches. Explored stability results for triangle factors leveraging equitable colouring theorems and absorption methods. Proved key connectivity lemmas enabling absorber constructions and established asymptotic stability results. Ongoing work aims to complete the absorption framework to extend these findings toward a broader conjecture on clique-factor robustness.

The Brittleness of Imputation: An Information-Theoretic Audit of Imputation Methods
Supervisor: Ioanna Papatsouma - Imperial College London,
June - August 2025

- **Overview**: This project investigates the extent to which widely used imputation methods preserve meaningful structure in datasets, particularly for downstream clustering tasks. While traditional evaluation metrics such as NRMSE and PFC quantify pointwise accuracy, they fail to capture whether the imputed data maintains the original information topology. Using an information-theoretic clustering method (DIBmix), we audit the "brittleness" of several imputation techniques—including MICE, FAMD, and MissForest—across multiple real-world datasets. Our findings reveal a frequent divergence (54%) between low imputation error and high structural fidelity, highlighting the need for evaluation frameworks that consider task-specific robustness rather than pointwise accuracy alone.

An Algebraic Solution of Transport Across the Retinal Pigment Epithelium
Supervisor: Allanah Neff - The University of Edinburgh,
May - October 2024

- **Overview**: Built on previous models (Dvoriashyna 2018) by substituting in more accurate non-linear processes like the Goldman-Hodgkin-Katz equation. Using scientific programming and techniques of non-dimensionalisation, we found that conductance had a large influence on the end behaviour of the relation between pump-rate and cell volume. A possible future route was to apply new experiment-based conductance data to refine the assumptions made in prior papers, the significance of conductance in our models suggests that this would notably improve accuracy. However, due to other time commitments the project for now is halted.

Selected Honors and Awards

2026 School of Mathematics Travel Fund - *The University of Edinburgh*
Awarded funding to attend ASIACOMB 2026, Daejon, South Korea.

Royal Society - Women & Future of Science Conference - *The Royal Society*
1/11 selected from In2research alumni to present my Imperial College research

2026 MITACS Globalink Research Internship - *University of Victoria*
Supervisor: Dr Jonathan Noel - Extremal Combinatorics
Competitive international research award for Summer 2026 in Canada

2025 In2research Programme Participant - *In2Science UK*
Supervisor: Dr Ioanna Papatsouma - Imperial College London
Fully funded 8-week research project in Machine Learning & Data Science

2025 Mathematics Vacation Scholarship - *The University of Edinburgh*
Selected, but ineligible due to the Imperial College London research placement

Ritangle Mathematics Competition Final Round Competitor - *MEI*

The Parkinson Memorial Prize - *Sir Roger Manwood's School (SRMS)*
For Year 13 Boys who have made an outstanding contribution to the life of the school

Year 13 Engage Award & **Year 12 Excel Award for Mathematics** - *SRMS*

Outreach & Engagement

UK Mathematics Trust, Leeds, UK, August 2024 & 2025
National Maths Summer School Mentor
- Taught an Olympiad primer on the pigeonhole principle for 35+ students from the top 1.5% of the Intermediate Maths Challenge.
- Assisted on 'Ramsey Theory' for British Math Olympiad Round 1 & 2 candidates. All funded by XTX Markets.

The University of Edinburgh, Edinburgh, UK, October 2023 - Present
School of Mathematics Outreach Team
- Náboj Mathematics Competition - Problem Exchanger
- Edinburgh Maths Circle - Problem Setter
- Edinburgh Science Festival 2024 - Knot Theory and Recursion Stall

The University of Edinburgh Maths Society, Edinburgh, UK, October 2024
Estimathon Organiser
- Coordinated and hosted an event with 50+ attendees.
- Brought together 5+ student bodies. All sponsored by Jane Street.